

Raunak Sood

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Education

UNIVERSITY OF SOUTHERN CALIFORNIA

Los Angeles, California

Master of Science in Computer Science

Expected: May 2027

- **GPA:** 3.85
- **Relevant Coursework:** Analysis of Algorithms, Applied Natural Language Processing, Database Systems, Deep Learning, Operating Systems
- **Projects:** leakprobe (open-source leakage testing, PyPI), Bias Fingerprints and cardinality-eval (sole-author papers), filing-links (point-in-time 10-K graph), Melange (creative collaboration app), Resampled Thought Trees (LLM reasoning)

SANTA CLARA UNIVERSITY

Santa Clara, California

Bachelor of Science in Economics

June 2024

- **GPA:** 3.9 (Upper-Division), Dean's List 2022-2024
- **Relevant Coursework:** Machine Learning, Time Series, Algorithms, Data Structures, Linear Algebra, Dynamic Macroeconomics

Experience

INNOVIUS CAPITAL

San Francisco, California

Machine Learning Intern

May 2026 - Present

- Replaced chance-level PCA company scoring with a supervised CatBoost + LLM ranker, raising P@20 0.20 to 0.90
- Built the 14k-company point-in-time training set from raw CRM data; a label-leakage fix raised AUC 0.62 to 0.66
- Shipped an unattended weekly retrain loop with drift triggers and promotion gates, scoring all 14k CRM companies
- Built a 268k-company embedding search engine (hybrid retrieval + LLM rerank) with 28/28 held-out recall

QIAGEN

Redwood City, California

Machine Learning Intern

June 2025 - Aug 2025

- Built an embedding-based reasoning system (GraphRAG) to explain disease target pathways to scientists
- Fine-tuned and deployed an LLM system to surface gene-variant information, reducing research time by 70%

QIAGEN

Dusseldorf, Germany

Data Science & Global Treasury Intern

July 2024 - Sept 2024

- Built a Langchain RAG chatbot to automate financial dashboard access, reducing internal query time by 50%
- Conducted FX analysis on \$10M+ transactions, monitoring currency risk and making trades on Bloomberg

YORK UNIVERSITY

Toronto, Canada

Data Science Intern

Sept 2023 - Dec 2023

- Applied topic modeling on earnings calls to explore incumbent investment firms attention to financial technology
- Performed regression analysis to find what factors drive discussion and adoption of technology within these firms

PRECANTO

Los Gatos, California

Data Science and Product Intern

July 2023 - Dec 2023

- Built and evaluated time series models for headcount forecasting (hiring/attrition) across several companies
- Automated validation of the predictive payroll tax engine, increasing overall validation efficiency by 90%

GREENFIELD RESEARCH PARTNERS

Palo Alto, California

Private Equity Research Associate

Feb 2023 - May 2023

- Sourced 100+ acquisition targets in the real estate sector, identifying 5 potential deals worth \$50M+
- Assisted in LBO modeling and comps analysis for exit multiple estimation

Research and Open Source

LEAKPROBE

Open source, pip install leakprobe

Sole author

Sep 2026 - Present

- Tests feature pipelines for temporal leakage with no ground truth: perturb each declared source and flag anything that moves
- Found a leak that inflated mean IC 59% in a 22-factor study; catches 8 of 9 planted leaks on five public datasets, no false positives

BIAS FINGERPRINTS

Sole-author paper, q-fin.ST preprint

Point-in-time factor pipeline, 22 factors over ~500 S&P names

2025 - 2026

- Proposes reading a published factor table backwards to identify the data-handling error it contains, from measured signatures
- Period-end dating inflates mean IC 59% and manufactures four spurious t-statistics; survivorship rearranges, and only it is diagnosable

CARDINALITY-EVAL AND FILING-LINKS

Sole-author paper and extraction pipeline

LLM evaluation and information extraction

2026

- Annotation-free instrument for extraction recall vs. item count; free-form recall falls 0.214 from 1 to 16 items, schema-constrained does not
- Extracted 10,382 verbatim-validated customer/supplier claims from 3,298 10-Ks with a local 30B model into a point-in-time graph

Leadership and Activities

USTA

Los Altos, California

Tennis Player

Jan 2010 - March 2021

- Ranked top 6 in Northern California and member of Santa Clara University's Division I Men's Tennis Team

Skills and Interests

Technical: Python, R, Excel, SQL, C++, Java, Docker, LATEX, Bloomberg, Jupyter, Tensorflow, Pytorch, LangChain, Git, AWS

Interests: Wildlife, Golf, Poker